EXPLORATORY DATA ANALYSIS OF THE 2019: NEW YORK CITY AIRBNB OPEN DATA

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Abstract

This exploratory dataset analysis project focuses on analyzing the 2019 New York City Open Data Airbnb dataset using Python. The dataset was loaded into Google Collab, then examined its structured and then it was cleaned to address any missing values and remove less useful information. Afterwards, exploratory data analysis (EDA) was conducted to identify tends and relationships between variables. Several statistical visualizations were created to present the data in an easily accessible and actionable format. This analysis aims to provide data-driven insights for Airbnb hosts and future hosts who want to optimize their listing strategy based on observed trends in location, pricing, and room type.

Introduction

New York is one of the most diverse and tourist places in the United States, making Airbnb listings highly attractive and a good business to dive into. The availability of Airbnb listings is influenced by a variety of factors including location, room type, and pricing.

The following is a variety of data driven analysis from a 2019 New York City Airbnb dataset. These visualizations present the geographic location of listings in New York City, the number of listings offered by room type, the price distribution of all locations, and the price variability by neighborhood groups such as: Brooklyn, Manhattan, Queens, Staten Island, and Bronx.

The analysis provided serves to assist both current and future Airbnb hosts make an informed decision that would align with their goals: whether it's maximizing profit, increasing levels of occupancy throughout the year, or help determine an ideal location to publish a new listing.

Methodology

The dataset was extracted from Kaggle.com under the name of "New York City Airbnb Open Data" and imported into Google Collab for inspection and analysis. After reviewing the dataset's cell information, checking for missing values, and duplicated entries, missing values were filled with zeros in column 'reviews_per_month" to assume that there were no reviews for a listing. Following this, the term 'Unknown" was input into missing values under "Name" and "host_name" columns to assume that there was no information about the host. Lastly, the "last_review" column was removed due to the lack of relevance to objective that this dataset aims to achieve.

After cleaning the dataset, the exploratory data analysis (EDA) techniques were put into play. A series for 5 data visualizations were generated to understand relationships between entities such as price, location, availability and room type.

NYC Geographic Map

To begin the data analysis, the geographic map of New York City (Fig. 1) was generated with a sample of the first 1000 Airbnb listings being color-coded by neighborhood groups to provide a visual comprehension of how much Airbnb's have populated the different neighborhoods.

Map Approach

The geographic map was generated by importing the folium package from Python. The map was centered on the latitude and longitude coordinates of the center of New York City, and tile layers were added for clearer visibility and providing a different map style while keeping the map layer than Folium generates. Due to the excessive amount of data in the dataset, the conscious decision of creating a map using sample data of the first 1,000 rows of data to provide further clarity in the map but also decrease the run time of the program. The 1,000 values were then plotted using Folium's circular markers and rearranged into their appropriate neighborhood groups by being color coded. Additionally, a legend was defined using html and provided at the top left-hand side to indicate each neighborhood group where: Manhattan is orange, Brooklyn is red, Queens is green, Bronx is purple, and Staten Island is dark blue. Lastly, the title was added to the top of the map, and the map was downloaded as a html file.

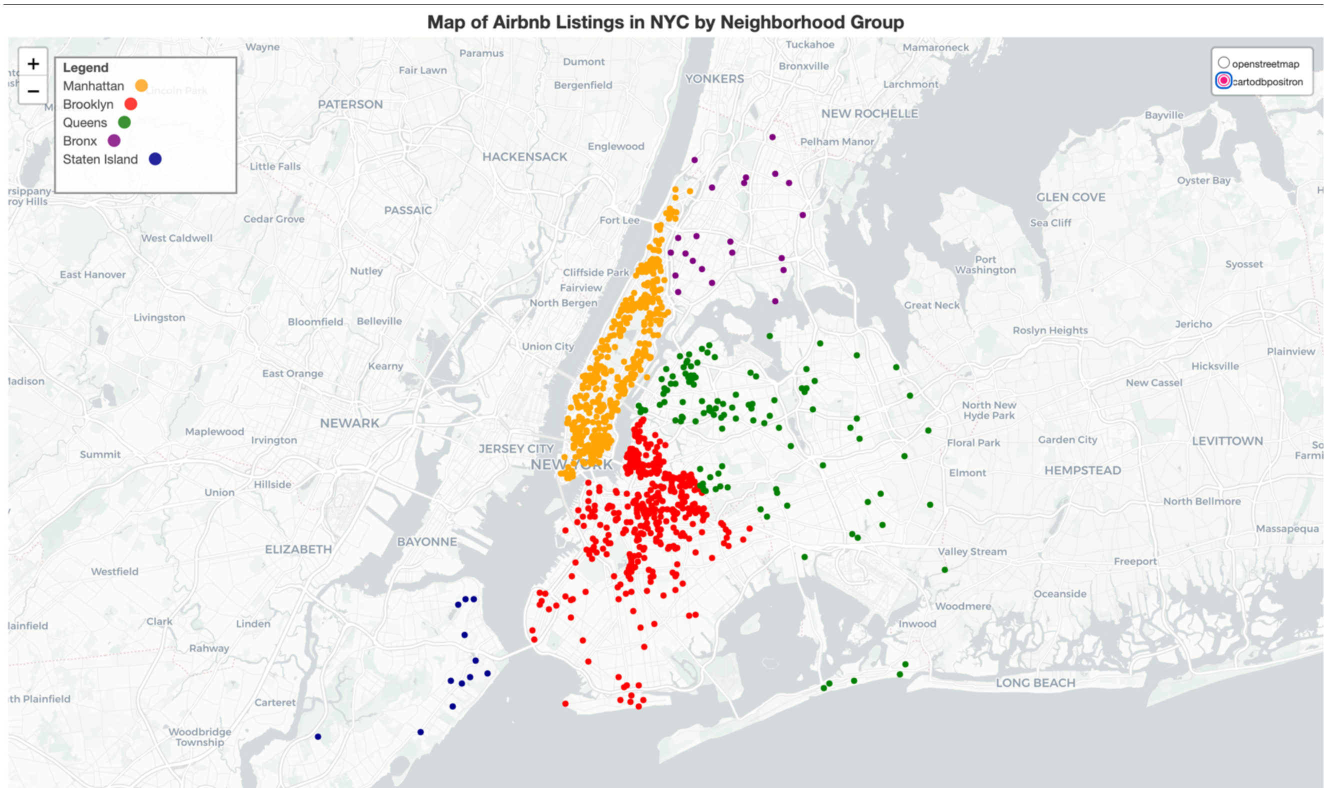

Figure 1

Map Description

The map shows fewer listings in Staten Island and Bronx and a higher Airbnb activity in areas like Manhattan and Brooklyn. This could indicate that there is a higher demand by tourists to stay in the Brooklyn and Manhattan sectors and perhaps there's more activity and more tourist attractions nearby. On the other hand, Staten Island and Bronx may have less tourist attractions and most housings are occupied by New York City residents.

As an Airbnb host or looking into getting into the market, this map assists as a guide to look into creating listings where there's much more activity and a higher demand or investigate listings where there is low competition and provides a stay for people that prefer calmer settings.

Price Distribution Across Listings in NYC

The price distribution of all listings in NYC is just as much of an important visual as the geographic map of NYC to price an Airbnb properly. A histogram of the price distribution of all the listings (fig. 2) best represents where the price range tends to fall for a night in NYC. This histogram doesn't dive into neighborhoods but rather provides a broad understanding of the price points and identifies any extreme values and outliers.

Pricing Distribution Using Python Libraries

Through importing python packages: Pandas, Seaborn and Matplotlib, we are able to generate and label a histogram to present the price distribution from the dataset using the column titled "price". From there, 100 bins were selected to more appropriately provide a clear yet detailed graphical understanding of the segments. Each bar represents intervals of 100 units (in this case, in dollars). The limit of the table was set to $2,500 on the x-axis to both increase the view of the graph and still identify extreme outliers. Lastly, the graph was labeled, titled and color coded.

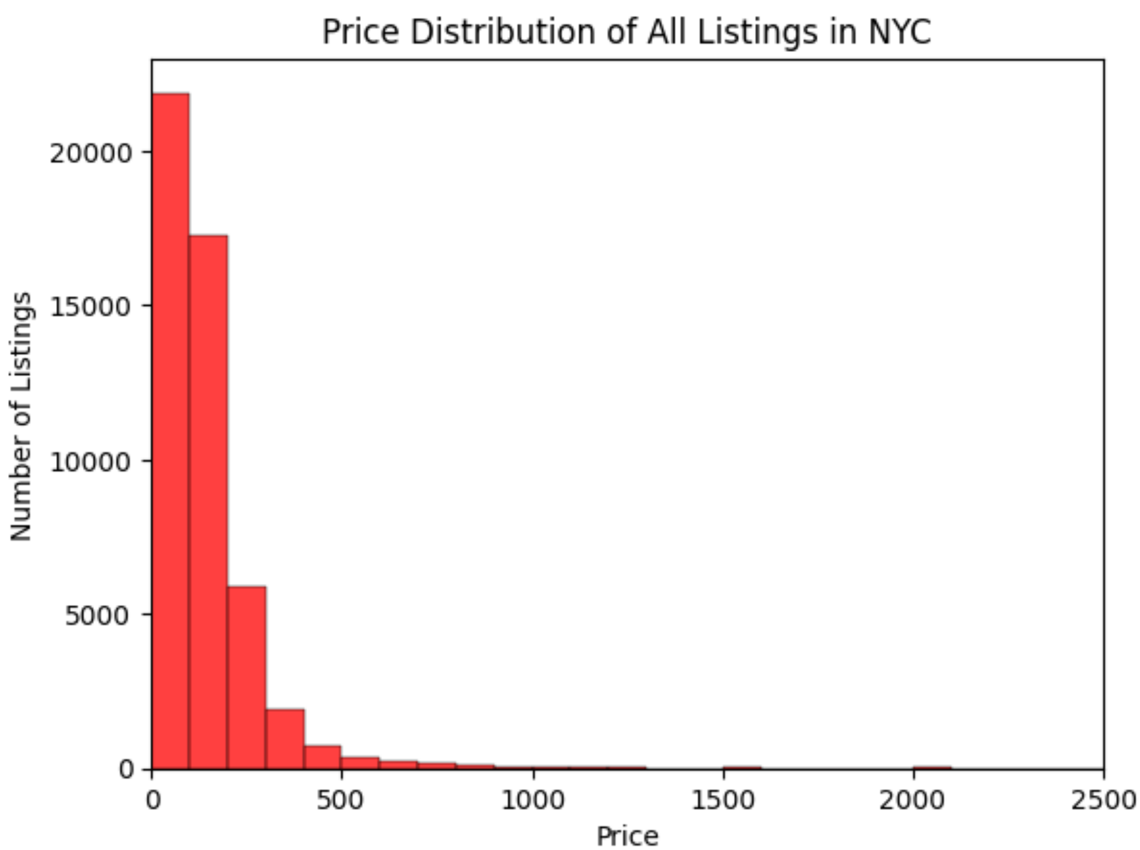

Figure 2

Price Distribution: Histogram Description

The histogram presents a distribution of the price range of all listings in NYC regardless of the location. The graph demonstrates a right skew and a sharp decline after $200 price ranges where Airbnb listings decrease but the price continues to increase. The right-skewed distribution also has a notably long tail where there are close to zero Airbnb listings available at prices over $2,000.

Based on the histogram, we can assume that most of the listings are below $200 a night and the long tail indicates that there are a few high-end Airbnb listings available in NYC but most listings fall under an affordable price range. This chart helps hosts price their listings accordingly and give them an idea of where other people are setting their prices. Through this report we can take other factors into consideration to have a better understanding of the price ranges and where a listing would thrive.

Price Range by Neighborhood Group

While in the topic of price ranges, we can narrow down specifics of a price range into Neighborhood groups. The visualization distinguishes the prices per neighborhood group and identifies outliers within each sector.

Analyzing Price Ranges by Neighborhood Groups Using Python

Packages pandas and matplotlib and seaborn were used to generate the cat plot, use the columns "neighbourhood_group" on the x-axis and "price' on the y-axis. Initially a box plot, violin plot and a categorical plot were created using the same information to compare a better graphic and the categorical plot best identifies the outliers and clusters where most of the prices fall for a specific neighborhood and graphically interprets the relationship of a categorical and a numerical value.

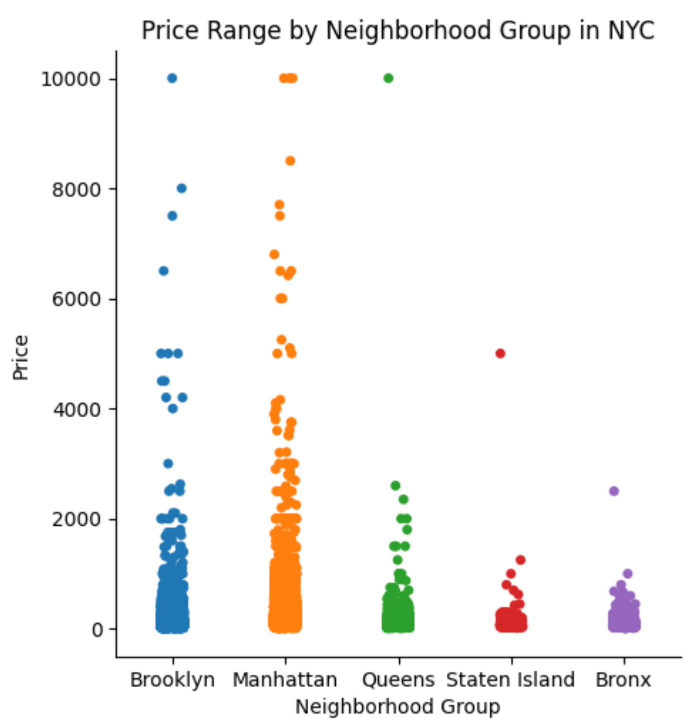

Figure 3

Interpreting the Categorical Plot

The categorical plot shows the price range of neighborhood groups where Manhattan shows the widest range of prices up to $10,000 per night listings. Brooklyn is a close second in price ranges but most of their price range falls under $2,000 with some notable outliers at $10,000. Bronx and Staten Island show a lower price range where values tend to be at around a maximum range of $1,000 per night with two outliers indicating an Airbnb that may be at a higher end. Queens differs from the other locations since there are no notable outliers, indicating that the price range is less versatile in this area and may not offer high-end luxury locations.

In conclusion, Manhattan and Brooklyn offer the widest range of accommodations for all tourist in different tax-brackets and the higher end accommodations may tend to thrive more in these areas since there's several listings at higher prices.

Do Room Types Matter?

For a host to make a well-informed decision of what type of room they wish to provide and what type of room would optimize profit due to the number of days that the Airbnb is occupied for throughout a 365-day period, a count plot and a boxplot were created to provide a visible comparison of supply and demand.

Creating Two Graphs to get to One Conclusion

For the count plot, data from the columns: "room_type" and "number_of_listings" was used to present the total number of listings offered in each type of room: private room, entire home/apt, and shared rooms. The graph was then color coded by room type for the viewer to distinguish the different categories.

The box plot analyzes the availability out of the year in days (365 days total) for each room type rather than using the total number of listings. In the x-axis we use "room_type" from the dataset and we use "availability_365" for the y-axis. The dataset is then color coded and used to present a comparison between the two graphs.

This approach to combine both graphs hopes to give the host an indicator of what Airbnb are available and what type of Airbnb is usually booked throughout the year and available the least number of days.

Interpretations and Inferences

In figure 4, "Number of Listings offered per Room Type in NYC" we can see that most of the Airbnb listings are entire homes or apartments with a listing count of 25,000 places. About 22,000 more listings are classified as private rooms and a notably few numbers of listings (less than 5,000) are shared room listings. Based on this table alone, we could infer that perhaps there is a higher demand for Airbnb that offers an entire home/apartment and private rooms rather than shared rooms or that a shared room is not something hosts have interest due to potential conflict.

Figure 5 shows the number of days that were available to book in the year of 2019. As seen below, private rooms and entire homes had a lower median range of being available for only 45 days out of the year by the end of 2019 in comparison to a shared room where it had a remaining availability of 100 days out of the year. We could create the assumption that private rooms and entire home/apt are booked first by visitors. In addition, shared rooms have a significantly wider range of values where the third quartile is found at around 340 days of the year or, in simpler terms, more than 75% of the shared room listings are available for most part of the year. More availability could signify that these are listings that people aren't inclined to or that there is a significantly higher amount of shared room listings and that explains why they are often, available.

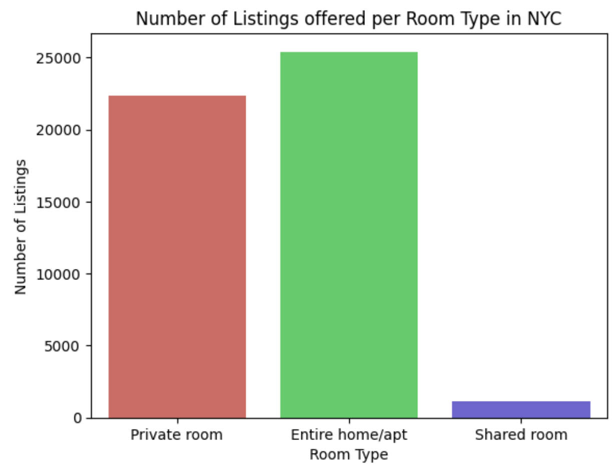

Figure 4

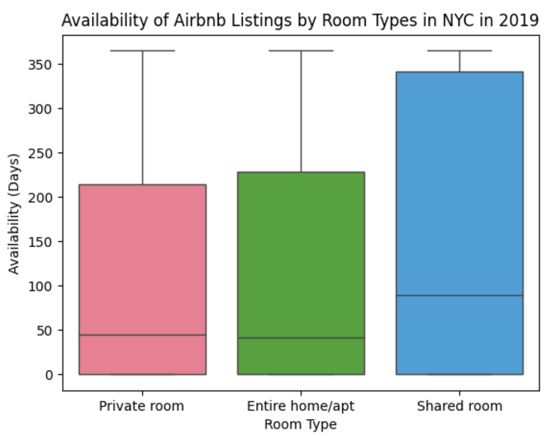

Figure 5

Using the data and knowledge of figure 4 and figure 5, a better interpretation of the room types can be assessed. As seen in figure 4, both private and entire home/apt room types have a higher number of listings but also have a lower availability left by the end of the year. The demand for private rooms is relatively very high for the supply of private rooms and entire homes. In contrast, there is very few listings for shared rooms and the very few shared rooms available are not being booked frequently. There is not the same level of demand for a shared room as there is for private rooms and for hosts, this would best be a type of room they would want to avoid listing if they want to maximize occupancy.

Conclusion

The EDA performed from using the 2019 New York City Open Data Airbnb dataset demonstrates the importance of observed trends using location, pricing and room type performances to initialize and optimize a listings success. The geographic map showed the 2019 listings populating differently in neighborhoods primarily occupying the Manhattan and Brooklyn area. The price distribution histogram provided a price range per night in NYC where most listings fall under the $200 price range and we sorted those prices into neighborhood groups to have an idea of what prices can you expect to pay in specific neighborhoods. Manhattan and Brooklyn provided the widest variety of price ranges indicating that these areas offer listings that are affordable and other listings that reflect a more luxurious stay. Lastly, we concluded that the type of room someone decides to list will impact your ability to generate profit as a private room and entire homes and apartments populate the market but they are also the more frequently occupied which in this case, trying to stand out from the market by providing a shared room wouldn't be the most optimal decision.

What's next?

As this report provides well-produced and insightful data, there is room to expand on more decision-making tools. For future research, more variables could be added on into what determines the demand for room types and how to price a room type. Using prices in relation to room types would better give the target audience an idea of how expensive they should price their listing and using these four combined variables: availability, price, room type and neighborhood group to form a pricing strategy.

Next, we could incorporate more variables such as number of reviews a listing gets per month and when the last review was to investigate what hosts are doing right and what people can do to increase their chances of continuous profit.

Furthermore, future research plans include creating machine learning models to predict occupancy rates based on a multitude of factors such as season, previous data, pricing across the market, and location.

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